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MILITARY SPECIFICATION

TRUCK, LIFT, AERIAL STORES 3000 POUNDS CAPACITY MJ-lA AND MJ-lB

* This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scone. This specification covers two types of self-motivated, power hydraulic lift trucks.

Type 1 - Powered by a 4 cylinder gasoline engine, MJ-lA
Type 2 - Powered by a 2 cylinder diesel engine, MJ-lB

* 2 APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in. that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitalion form a part of this specification to the extent specified herein

SPECIFICATION

FEDERAL

VV-F-800	Fuel, diesel

MILITARY

MIL-C-104	Crates, Wood: Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-L-2104	Lubricating Oil, Tnternal Combustion Engine, Tactical Service.
MIL-L-2105	Lubricating Oil, Gear, Multipurpose.
MIL-G-3056	Gasuline Automotive, Combat.
MIL-F-3541	Fittings, Lubrication.
MIL-G-5514	Gland Design, Packings, Hydraulic, General Requirements for.
MIL-H-5606	Hydraulic Fluid, Petroleum Ease, Aircraft, Missile, and Ordnance.
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, TX 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1730

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MI L-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specifications for.
MI L-G-10924	Grease , Automotive and Artillery”
MI L-B-11188	Batteries, Storage: Lead Acid.
MI L-R-12144	Reflector, Indicating, Clearance,
MI L-G-23827	Crease, Aircraft and Instruments Gear and Actuator Screw.
MIL-I-45208	Inspection System Requirements.
MIL-T-83133	Turbine Fuel, Aviation, Kerostne Type. Grade Jp-8

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes .
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	identification Marking of U.S. Military Property.
MIL-STD-143	Standards and Specifl cations, Order of Precedence for the Selection of
MIL-STD-162	Materials Handling Equipment, Preparation for. Shipment , Storage, Cyclic Maintenance, Routine Testing and Process.
MIL-STD-808	Finish, Materials and Processes for Corrosion Prevention and Control in Support Equipment.
MIL-STD-810	Environmental Test Methods and Engineering, Guidelines,
MIL-STD-831	Test Reports Preparation of.
MIL-STD-889	Dissimilar Metals.

* (Coples of specifications and standards required by manufacturers in connection with specific acquistion functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.1.2 Other Government documents. drawings, and publications . The following other Government documents form a part of this specification to the extent specified herein.

DRAWINGS

AIR FORCE

65D35518	Specification, Welding.
69E51584	Assembly, Engine
7447187	Truck, Lift, Aerial Stores, 3000 Pounds Capacity, MJ-1A , Assembly of,
8141033	Engine, Source Control Drawing.
8141050	Truck, Lift Aerial Stores, 3000 Pounds Capacity, MJ-lB, Assembly of.
7545352	Requirements for Finishes, Protective, and Codes for San Antonio ALC Ground Support Equipment.

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UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, IL 60606.)

* 2.1.3 Order of precedence. ln the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence .

3. REQUIREMENTS

3.1 First article testing. This specification makes provision for first article testing.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities services not specitied herein shall be selected in accordance with MIL-STD-143 except as provided in 3.2.1 and 3.2.2.

3.2.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are not suitable standard parts. in any case, commercial utility parts, like screws, bolts, nuts, and cotter pins, having suitable properties may be used provided:

a. They can be replace by standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical on the drawings

3.2.2 Standard parts. With the exception in 3.2.1, MS and AN standard parts shall be used where thtey suit the purpose. They shall be identified on the drawings by the part number.

3.3 Materials.

3.3.1 Metals. Metals shall be of the corrosion-resistan type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric condition likely to be met in storage or normal service. When materials are used in the construction of the lift truck that are subject to corrosion, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

3.3.2 Dissimilar metals. Unless suitably protected against electrolytic corrion,dissimilar metals shall not be user in intimate contact with each other Dissimilar metal, are defined in MIL-STD-889

3.4 Design and construction. The design and construction of the type 1 lift truck shall conform to drawing number 7447187. The design and construction of the type 2 lift truck shall conform to drawing number 8141050.

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3.4.1 Maintenance. The lift truck shall have a minimum number of parts consistent with reliability. The design shall permit easy assembly and maintenance with tools normally available commercially by service maintenance personnel with a minimum of training.

3.4.2 Screw threads, Screws threads shall be in accordance with MIL-S-7742

3.4.3 Hydraulic and hydrostatic fluid. The fluid used in the hydraulic-hydrostatic system shall be automatic transmission fluid, type A, suffix A.

3.4.4 Lubrication. Greases for general chassis lubrication shall conform to MIL-G-10924. All grease fittings shall be accessible for pressure type lubrication without need for disassembley at any part of the vehicle. Grease for lubrication of antifriction bearings shall conform to MIL-G-23827.

3.4.4.1 Engine oil. Engine oil shall conform to MIL-L-2104.

3.4.4.2 Gear lubricant. Gear lubricant shall conform to MIL-L-2105.

3.4.4.3 Grease fittings . Grease fittings shall be in accordance with MIL-F-3541 .

3.4.5 Structural strength. The structural strength of the lift truck shall be such that there will be no mechanical failure or deformation of any welds or components as a result of tests described in 4.6.

3.5 Performance. The lift truck shall be capable of performing the picking un,, transportation and positioning operations necessary for lifting aerial stores with weights up to 3000 pounds to the racks and pylons of aircraft. The lift truck shall be capable of performing such functions under the conditions described in section 4.

3.5.1 Lift. The empty lift truck shall be capable of a reach of 75 inches, measuring from the ground to the top ant] center of the lift cradle.

3.5.1.1 Lift rate. The lifting time of the hydraulic system cantilever arms , up or down, fully loaded or empty, shall be not more than 12 seconds for the maximum lift of O to 75 inches. The lift shall be fully controlled in increments of 1/8 inch from minimum to maximum lift position, up or down, under a capacity load.

3.5.2 Speed The loaded lift truck shall have a speed of not less than 10 miles per hour on a hard level surface when the engine is operating at the speeds specified in 3.6.2.

3.5.3 Traction. The lift truck shall be capable of climbing a 20 percent grade empty, and a 10 percent grade loaded, without stalling

3.5.4 Transport handling The lift truck shall be provided with dual purpose tie-down rings. These rings shall be of sufficient strength, and so located, as to properly hold the lift truck in place during air transportation, and shall be suitable as lift points for lifting by crane.

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3.5.5 Engine loading. The fully loaded lift truck shall operate at maximum forward speed on a flat level surface and shall be capable of simultaneous operation of the lift arm and the steering system without stalling the engine.

3.5.6 Safety provision. Mechanical, hydraulic, and hydrostatic systems shall incorporate inherent safety features and provisions to protect operating personnel , equipment, and aircraft.

3.5.6.1 Failsafe. The lifting mechanism shall be equipped with a hydraulically actuated “Failsafe” device which prevents the lift arms from falling in the event of hydraulic failure.

3.6 Major components.

3.6.1 Power train, The power train shall consist of those units necessary for the application of generated engine power to the drive wheels of the lift truck. When lnstalled in the lift truck, it shall be-capable of satisfactory operation in any ambient temperature range from -40° Fahrenheit (F) to +125°F. It shall withstand, without damage, storage temperatures of -80°F to +180°F. The lift truck shall be equipped with a hydrostatic drive train which includes all the components and parts essential for adequate and efficient transmission of power from the engine to the differential under all conditions specified herein. The hydrostatic system shall be a closed loop hydraulic system containing a pump driving motor. heat exchanger and filter. This system shall also contain a mechanically actuated rotary by-pass valve to allow the lift truck to be towed in the event of engine/battery failure. In the open position, the valve shall by-pass the hydrostatic fluid, thereby removing the resistance of the hydrostatic system from the drive wheels. A foot operated pedal shall provide a variable speed control for the lift truck. The speed controls shall also provide inching capability both in the forward and reverse directions for precise positioning of the load. The speed control shall be a single acting foot pedal. This pedal shall cause forward, reverse, or zero movement of the lift truck when depressed while the direction selector handle is in either forward, reverse, or neutral, as applicable. Downward pressure on the pedal shall cause movement of the hydrostatic pump swash plate. The lift truck shall be equipped with a dash mounted forward, reverse, neutral (park) selector handle which selects the direction the lift truck will travel when the foot pedal is depressed. This handle shall include a microswitch feature to prevent the engine from being started unless the handle is in the neutral (park) position, Movement of the selector handle shall not cause movement of the hydrostatic pump swash, plate. The selector handle shall be designed to positively remain in the selected position yet allow the driver to easily change the selector from one position to another. With the selector in any position inadvertent movement of the lift truck in any other direction shall be prevented. A feature shall be provided to prevent the direction selection handle from being moved (shifted) when the foot pedal is depressed. The transmission and transmission controls shall provide for a smooth and positive destroking of the pump when the foot pedal is released,

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3.6.1.1 Heat exchanger. The heat exchanger on the lift truck shall limit the hydrostatic fluid temperature to +200 F with the lift truck loaded to capacity and traveling at maximum speed on a smooth level surface. This temperature requirement applies to non-continuous (normal) operation of the lift truck. During the continuous operation encountered in road testing (see 4.6.7), the maximum allowable temperature shall be +210°F.

3.6.2 Engine. The lift truck will be powered by an air cooled engine. The type 1 lift truck will have a gasoline engine conforming to USAF Drawing No. 69E51584. The type 2 lift truck will be powered by a diesel engine conforming to USAF Drawing No. 8141033. The operating speed of the engine shall be :

Type 1 - 2400/2550 RPM
Type 2 - 2500/2650 RPM

idle speed for both engines shall be 1100/1300 RPM.

3.6.2.1 Fuel. The type 1 lift truck shall operate on 73 octane gasoline per MIL-G-3056. The type 2 lift truck shall operate on diesel per VV-F-800, or turbine fuels, grades JP-4 or JP-5 per MIL-T-5624 and grade JP-8 per MIL-T-83133.

3.6.2.1.1 Test fuels. The environmental tests, paragraph 4.6.3, and the road tests, paragraph 4.6.7. shall be conducted using JP-4 fuel. Other tests shall use commercial DF-2 diesel fuel, TP-4 fuel is not readily available on the commercial market. The supplier is responsible for coordination JP-4 fuel purchase, through the procuring activity, from the nearest Air Force installation and the associated bulk fuel transfer and storage.

3.6.3 Electrical system. The type 1 lift truck shall have a 24 volt electrical system It shall incorporate an alternator system and a voltage regulator. The alternator shall produce sufficient power to prevent battery discharge when the engine is at operating speed and with the vehicle lights turned on. The type 2 lift truck shall have a 12 volt electrical system. It shall incorporate an internal flywheel type alternator and a voltage regulator. The alternator shall produce sufficient power to prevent battery discharge when the engine is at operating speed and with the vehicle lights turned-on. At maximum engine operating speed (3.6.2), the alternator shall provide not less than 10 amps.

3.6.3.1 Starting system. The starting system shall include a lead acid battery, starting motor, and starting switch. The type 1 lift truck shall incorporate a battery conforming to MIL-B-11188. The type 2 lift truck shall Incorporate a 75 amp-hour (minimum) battery conforming to SAE Group 24. As an aid in cold starting the type 2 lift truck shall be equipped with two manifold heaters and a switch mounted on the instrument panel for activation of the heaters. The battery shall have sufficient capacity to start the engine when tested. in accordance with 4.6.3.2.

3.6.4 Lighting system The lift truck shall be provided with a spotlight, a backup ligiht, and reflectors. Reflectors shall be in accordance with MIL-R-12144

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3.6.5 Chassis frame. The chassis frame shall be rigidly constructed with a capability of withstanding, without mechanical failure, the tests described in 4.6.

3.6.6 Steering. Steering of the two rear wheels shall be accomplished by a power steering mechanism. The steering gear ratio shall insure positive control of the lift truck while traveling at maximum speeds and be capable of turning the wheels while the lift truck is stationary.

3.6.7 Brakes. The brakes shall hold the loaded lift truck on a twenty percent grade.

3.6.7.1 Service brake. The lift truck shall have hydraulic brakes. The brakes shall be foot pedal controlled and capable of stopping the fully loaded lift truck traveling at 10 mph within a maximum distance of 16 feet.

3.6.7.2 Parking brake. The lift truck shall be equipped with hydraulic parking brakes. This shall be provided by locking hydraulic pressure in the service brakes. A valve will be provided in the operator's compartment for this purpose. The parking brakes shall be capable of holding the loaded lift truck on a 20 percent incline when the rotary. by-pass valve is in the open position.

3.6.8 Hydraulic lift system. The lift truck shall be provided with a lifting capability of 3000 pounds, consisting of a complete hydraulic system, cantilever mechanist., lift arms and cradle, The control valves and hydraulic system design shall minimize hydraulic actuator drift or movement unless commanded by the operator. Actuator drift shall not exceed 1/2 inch per hour.

3.6.8.l Control valves. The hydraulic system shall 1 have' control valves for positive metering of the hydraulic fluid to insure a smooth operation. The lift truck shall be provided with two sets of control valves. one set, located at the operator's compartment. shall control lift The second set, located on the cradle, shall control lift, tilt, lateral and longitudinaal table motions.

3.6.8.2 Packings. Packing installation shall be in accordance with MIL-G-5514.

3.6.8.3 Pressure compensated system pump The lift shall be provided with a pressure compensated system pump. This pump shall automatically vary its volume as required by demand on the hydraulic system. operating pressure shall be maintained in tile pump line only until a demand is made on the hydraulic system. Maximum operating pressure shall be 1500 psi.

3.6.8.4 Hand pump. The hydraulic system shall contain a hand pump which will provide power for lift motion in the event of engine or system pump failure.

3.6.9 Lifting mechanism. The lifting mechanisn shall be such that stores being listed shall remain in a given attitude during the lifting operation. The lifting, mechanism shall have a capacity of lifting stores from 6 3/8 inches above ground leve1 to 75 inches minimum above ground level. Provisions shall be made to prevent binding of the lifting mechanism while lifting with the lift truck setting on an uneven surface.

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3.6.9.1 Lifting arms. The lifting arms shall be of sufficient strength to prevent tilting of stores during adjustment operations fore, aft, or laterally.

3.6.10 Load transfer. Provisions shall be made for transfer of the weight of the loaded lifting arms to the main frame for relieving the hydraulic system while the load is being transported.

3.6.11 Instrument panel. The lift truck shall be provided with an instrument panel which will contain all switches, gages, etc. , required for normal starting and operation,

3.6.12 Cradle. The lift truck lifting arms shall be provided with an adjustable cradle for nesting and positioning the stores. The cradle shall be so designed that it will always be lower than the underside of stores, during loading and unloading operations. This cradle shall be equipped in such a manner as to nave four separate motions.

3.6.12.1 Yaw. The cradle shall impart to the weapon or store a yaw movement of 360 degrees minimum to either side of center. The yaw movement shall be manually operated.

3.6.12.2 Tilt. The cradle shall impart to tile weapon or store a tilt movement of 5 degrees minimum nose up or nose down. The tilt motion shall be hydraulically powered.

3.6.12.3 Longitudinal. The cradle shall impart to the weapon or store a longitudinal movement of 2 3/8 inches minimum fore or aft of center. The longitudinal motion shall be hydraulically powered.

3.6.12.4 Lateral. The cradle shall impart to the weapon or store a lateral movement of 2 3/8 inches minimum right or left of center. Lateial motion shall by hydraulically powered.

3.6.12.5 Cradle adjustment controls. Controls for adjustment of tilt, lateral, longitudinal , and of raising and lowering of the cradle shall be located on the cradle bed. Yaw adjustments shall be accomplished by manually pivoting the cradle on the bed.

3.6.12.6 Cradle rollers. The cradle shall be provided with adjustable rollers for nesting stores of different diameters, in accordance with the applicable drawing. The holes will be coded to conform to the different diameter requirements.

3.6.12.7 Tie-down strap. The cradle shall be provided with a quick connecting and disconnecting nylon tie-down strap with sufficient capacity to hold stores to the cradle during transportation.

3.7 Interchangeability. All parts having the same part number shall be functionally and dimensionally Interchangeable.

3.8 Treatment and painting. Treatment and painting shall be in accordance with MIL-STD-808 in conjunction with USAF Drawing No. 7545352. Color shall be No. 24052 per FED-STD-595.

3.9 Operation markings. Functional and operational controls shall be properly indicated and coded.

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3.10 Identification of product. Equipment , assemblies and parts shall be marked for identification in accordance with MIL-STD-130.

3.11 Workmanship.

3.11.1 General. The lift truck, including all parts and accessories, shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to the following;

a. Freedom from blemishes, defects, burrs, and sharp edges.

b. Accuracy of dimensions , radii of fillets and marking of parts and assemblies .

co Thoroughness of finishing, welding, painting, machining and soldering.

d. Alignment of parts and class fit of assembly, screws, and bolts.

3.11.2 Cleaning. The lift truck shall be thoroughly cleaned. Loose, spattered, or excess slag, metal chips or other foreign material shall be removed prior to subsequent production line operations.

3.11.3 Welding. Welding shall conform to USAF Drawing No. 65D35518.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of inspection requirements as specified herein, Except as otherwise specified in the contract, the contract may use his own or any other facilities suitable for the performance> of the inspection requirements herein, unless disapproved by the Government . The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the lift truck shall consist of the following.

4.2.1 First article tests. The first article tests are those tests performed on a first article lift truck to determine if that lift truck meets the requirements of this specification. First article tests consist of all tests described under 4.6.

4.2.1.1 Test sample. The first article lift truck, as specified in the contract, snail be tested for approval by the procuring activity, or when so specified in the contract, at the contractor's plant under the supervision of the procuring activity.

4.2.2 Acceptance tests. Acceptance test are those tests performed on production unitswhich have been submitted for acceptance by the Government. They consist of those tests described in 4.2.2.1 and 4.2.2.2,

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4.2.2.1 Production tests. Production tests shall be performed on each lift truck and consist of ail tests described below.

a. Cycle test with a 3000 pound load on the cradle, five complete cycles of each hydraulic motion.

b. Road test with a 3000 pound load on the cradle, one mile.

c. Service brake test per 4.6.9 except the grade shall be 10%.

d. Parking brake test per 4.6.10 except the grade shall be 10%

e. Structural overload test per 4.6.4 except load shall be 6000 pounds.

4.2.2.2 Periodic sampling tests. One lift truck shall be selected from each 50, or fraction thereof, on the contract or order and subjected to sampling tests. The test sample shall be selected by the procuring activity from production lift trucks which have successfully completed all production tests. The sampling tests shall consist of all tests described under 4.6 except for 4.6.3, environmental tests, and 4.6 .6, towing tests. The structural overload test (4.6.4) shall be performed at a test load of 6000 pounds.

4.3 Rejection and retest. When one item selected from a production run fails to meet the specification, no Item still on hand or later produced shall be accepted until the extent and cause of failure are determined.

4.3.1 Individual tests may continue For operational reasons, individual tests may be continued pending the investigation of a sampling test failure, but final acceptance of items on hand or produced later shall not be made until it is determined that items meet all other requirements of the specification.

4.3.2 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise’ the procuring activity of all defects likely to be found and methods of correcting them.

4.4 Test conditions. Unless otherwise specified herein, tests shall be performed under ambient atmospheric conditions prevailing at time of test.

4.4.1 Order of tests, The environmental tests (4.6.3) shall be performed first in any order convenient to the contractor and acceptable to the Government . Environmental tests shall be followed by structural overload tests (4.6.4). The remainder of the tests may be performed in any order convenient to the contractor and acceptable to the Government.

4.5 Test report. The contractor shall prepare a first article test report in accordance with MIL-STD-831 and furnish three complete copies of the report to the procuring activity.

4.6 Test methods.

4.6.1 Examination of product. Each lift truck shall be examined for conformance with this-specification and applicable drawing with respect to workmanship , material Dimensions, and markings.

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4.6.2 Functional tests. Each completely assembled lift truck shall be inspected for functional coordination of the power train, hydraulic system, lifting mechanism, and controls. Each truck shall be loaded to capacity and driven a minimum of 1/2 mile to assist in demonstrating the above.

4.6.3 Environmental testing. The lift truck, with a 3000 pound load on the table, will be subjected to the following environmental tests in accordance with MIL-STD-81O. In addition to the specific pass/fail criteria in the following paragraphs, a general criteria shall be that the lift truck complete the remainder of the first article tests without failure which is directly attributable to tile environmental tests.

4.6.3.1 High temperature test, method 501.l procedure I. The lift truck shall be operated at +125°F for 30 minutes by cycling the lift arms up/down The hydraulic oil temperature shall be monitored and shall not exceed +200°F.

4.6.3.2 Low temperature test, method 502. 1, procedure I The storage temperature shall be -65°F, The lift truck shall be operated at -40°F. Prior to starting the test, the vehicle shall be winterized as specified below and the battery shall be removed from the vehicle.

a. Drain, flush and refill hydraulic system with hydraulic fluid per MIL-II-5606

b. Drain engine crankcase and refill with engine oil per MIL-L-2104, grade 5W-20,

e. Drain differential and refill with oil per MIL-L-2105, grade 75W

4.6.3.2.1 Starting aids. To assist in cold starting of the engine, it is permissible to use two batterics connected in parallel, ether and a portable hot all blower.

4.6.3.2.2 Test conduct. After the engine has started, allow to run at 2000 to 2400 rpm for a period of not less than 15 minutes. Operate each hydraulic motion through complete cycles as specified below and in the order specified.

a. Lift - raise lift arms, pause for approximately 5 minutes. Lower lift arms. kaise and lower again to complete second cycle.

b. Actuate lateral, longitudinal and tilt motions through five cycles.

c. Operate steering through two complete cycles.

d. Operate drive forward and backward, as permitted by test chamber size,

e. Apply 3000 pound load to table and cycle each motion through two complete cycles.

NOTE: Physical size of the test chamber may prohibit performing a complete cycle of some motions. In this case a partial cycle- is acceptable.

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4.6.3.3 Humidity test. The lift truck shall be subjected to humidity tests in accordance with MIL-STD-810, method 507.1, procedure II After completion of the test, the lift truck shall be inspected. Some rust and corrosion is to be expected as a result of this test. The pass/fail criteria is as follows:

a. The lift truck shall be operated for 30 minutes at step 8 of procedure II by cycling each motion as permitted by the physical size of the test chamber. There shall be no evidence of abnormal operation.

b. The vehicle must be capable of completing the remainder of the first article tests without incurring a failure which is directly connected with such rust and/or corrosion.

c. There must not be patterns of rust and/or corrosion which may indicate that the contractor's manufacturing procedures are inadequate. Examples of such patterns are:

(1) General and gross flaking or peeling of paint.

(2) General and gross corrosion of hardware such as nuts, bolts, etc.

(3) Extensive corroision of an individual functional component such as a pump, motor or valve which prevents the component from operating or can reasonably be expected to prevent normal operation in the near future.

4.6.3.4 Sand and dust test, method 510.1, procedure I. The lift truck shall be operated for 30 minutes by cycling each motion as permitted by the physical size of the test chamber at step 5 of procedure I.

4.6.4 Structural overload tests. The lift truck shall be subjected to a static load test. The forward frames shall be blocked to prevent transmission of the load through the tires. The rear of the lift truck shall be chained or otherwise restrained to allow application of the full load. The lifting arm will be set so that the cradle is 30-50 inches of the ground. A 7500 pound load shall be placed on the cradle and held in position for 3 minutes. Evidence of yield in any of the components which could in any manner prevent the lift truck from meeting operational requirements during service life shall be reason to consider that the lift truck has failed this test.

4.6.5 Engine loading test. The lift truck shall be operated at maximum forward speed on a level hard surface road. Simultaneous operation of lift arms and steering shall not cause the engine to stall.

4.6.6 Towing test. With the engine off and the rotary by-pass valve in the open position, the unloaded lift truck shall be pushed 30 feet by two men without undue resistance of the drive train The truck shall then be loaded and towed .3 miles. Towing shall have no adverse effect on the hydrostatic drive system as evidenced by examination and successful completion of the remaining tests.

4.6.7 Road tests. The lift truck shall be subjected to a road test of 150 miles over paved surface, 75 miles loaded and 75 miles empty. At the end of this test, the lift truck shall be examined for construction deficiencies such as loose bolts and nuts, broken welds, and yields. For periodic sampling tests, reduce road test to 50 miles, 25 miles loaded and 25 miles empty. The temperature of the hydrostatic fluid shall be monitored and shall not exceed +210°F.

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4.6.8 Cycle tests. The lift truck shall be subjected to a cycle test which consists of raising and lowering a capacity load through the full lift range of the lifting mechanism for 1000 cycles. Cradle adjustments shall be actuated, one adjustment to each five lifting cycles. At the end of this test, evidence of Yield in any components that could in any manner prevent the lift truck from meeting operational requirements during service life shall be reason to consider that the lift truck has failed to meet this test. For the periodic sampling test, reduce test to 100 cycles.

4.6.9 Service brake test. The lift truck shall be placed on a 20 percent incline and loaded to capacity. The service brakes shall have sufficient capacity to hold the vehicle without movement. in addition, the service brakes shall stop the loaded lift truck traveling at 10 mph within a maximum distance of 16 feet.

4.6.10 Parking brake test. The lift truck shall be placed on a 20 percent incline and loaded to capacity. Application of the parking brake shall hold the vehicle without movement. The rotary by-pass valve shall be in the open positiorn for this test.

4.6.11 Attitude test. The lift truck cradle shall be subjected to a bomb attitude adjustment test with a 3000 pound simulated bomb mounted on its axis. The nose of the bomb shall be tilt nose down 5 degrees and up 5 degrees from horizontal and shall shift laterally 2 3/8 inches either side of center and longitudinally 2 3/8 inches fore and aft from the center of the bomb table. The bomb shall rotate about the vertical axis of the bomb table through 360 degrees.

4.6.13 Failsafe test. The cradle shall be raised off the ground. It shall be demonstrated by causing a hydraulic failure that the “Failsafe” features of the main lift mechanisms are functioning. The failure shall be simulated by disconnecting the return hydraulic line (line to rod end of cylinder) from one lift cylinder. The lime shall be disconnected at the counterbalance valve. There will be a “shifting” of the load as a result of disconnecting one lift cylinder. However, after this initial “shift,” there shall be no visible drop In the lift arms.

4.6.13 Manual lift test. The auxiliary hand pump shall demonstrate that the lift can be raise or lowered through its range with the engine shut off.

4.6.14 Lift rate. With the engine at maximum operating speed (3.6.2), the cradle shall be raised from, ground to maximum height. It shall not require more than 12 seconds to complete the lift cycle with a 3000 pound load.

4.6.15 Grade test. The loaded lift truck shall be stopped on a 10 percent s lope . From this position, the lift truck shall climb the slope. This test shall bc reptated on a 20 percent slope with tile lift truck unloaded.

4.6.16 Speed test With the engine at maximum operating speed (3.6.2), the lift truck. small demonstrated a minimum speed of 10 mph loaded.

4.6.17 Drive control test. It shall be demonstrated that the lift truck is cable of smoothly accelerating from 0 to 10 mph without evidence of “leaping” or "jumping." It shall be demonstrated that, from 10 mph, the lift truck will come to a smooth controlled stop when the foot pedal is released. This test will be performed on a hard level surface.

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4.6.18 Drift test. This test shall demonstrate that, with the engine operating at 1800 rpm and a 3000 pound load on the table, the hydraulic actuator drift does not exceed 1/2 inch per hour. Lift arm drift shall be measured at the front of the cradle. Tilt, longitudinal and lateral cylinder drift shall be measured relative to the respective cylinder rod,

4.7 Quality program.

4.7.1 Inspection system. Unless otherwise specified in the contract or purchase order, the contractor shall implement and maintain an inspection system in accordance with MIL-T-45208.

4.7.1 Sampling plan. Sampling plans shall be in accordance with MIL-STD-l05.

4,7.3 Level of inspection. Unless otherwise specified in the contract or purchase order, the inspection level shall be level II with an AQL of 1.5.

5. PACKAGING

5.1 Levels of preservation and packaging.

5.1.1 Level A. The lift truck shall be preserved and packaged in accordance with MIL-STD-162.

5.1.2 Level C. The life truck shall be preserved and packaged in accordante with manufacturer's commercial practice. The lift truck shall be prepared for shipment in such a manner as to assure adequate protection against corrosion and deterioration during shipment to the using, activity for immediate use

5.2 Levels of packing

5.2.1 Level A. The lift truck shall be crated to conform to MIL-C-104, style B, sillbase crating.

5.2.2 Level B. The lift truck, preserved and packaged in accordance with 5.1.1, shall he prepared for shipment as a mobile load.

5.2.3 Level C. The lift truck shall be prepared for shipment in such a manner as to insure safe transportation at the lowest rate to the point of delivery. The lift truck, so prepared, shall meet uniform classification rules or regulations of other carriers as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking specified by the contract or purchase order, the crates shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows. Truck Lift Aerial Stores, 3000 pounds capacity, MJ-1A or MJ-lB, as applicable.

6. NOTES

6.1 Intended use The lift truck covered by this specification is Intended for use in or around hardstands and landing strips, to pick up, transport, position and load aerial stores into aircraft

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6.2 Ordering data. Documents should specify the following:

a. Title, number, and date of this specification.

b. Level of preservation, packaging, and packing (see 5.1 and 5.2).

* 6.3 Changes from previous. The margins of this document arc marked with asterisks to indicate where changes (additions, modifications, corrections , deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractor's are cautioned to evaluate the requirements of this document based on the entire content, irrespective of the marginal notations and relationship to the last previous issue.

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Preparing activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions — Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-9720F(USAF)	TRUCK, LIFT, AERIAL STORES, 3000 POUNDS CAPACITY MJ-1A and MJ-1B

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* — Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* — Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* — Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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